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Chapter 6

Digital control systems

6.1 Sampled-data systems

6.1.1 Introduction

The control strategies that have been presented before were described by Laplace transform which are assumed to be built using analog electronics. However, most control systems today use digital computers (usually microprocessors) to implement the controllers. The use of a digital computer as a compensator (controller) device has grown during the past two decades as the price and reliability of digital computers has improved, (Franklin et al., 2006; Dorf and Bishop, 2008). The intent of this chapter is to expand on the design of control systems that will be implemented in a digital computer.

Figure 6.1(a) shows the topology of the typical continuous system. The computation of the error signal e and the dynamic compensation can all be accomplished in a digital computer as shown in Figure 6.1(b). The fundamental differences between the two implementations are that the digital system operates on samples of the sensed plant output rather than on the continuous signal and that the control provided by the compensator must be generated by algebraic recursive equations, (Franklin et al., 2006).

We consider first the action of the analog-to-digital (A/D) converter on a signal. This device samples a physical variable, most commonly an electrical voltage, and converts it into a binary number that usually consists of 10 to 16 bits. Conversion of the measured output from the analog signal $m(t)$ to the samples, $m(kT)$, occurs repeatedly at instants of time T seconds apart. T is the sample period, and $1/T$ is the sample rate in Hertz.

The sampled signal is $m(kT)$, where k can take on any integer value. It is often written simply as $m(k)$. We call this type of variable a discrete signal to distinguish it from a continuous signal such as $m(t)$, which changes continuously in time. A system having both discrete and continuous signals is called a sampled data system.

Usually, the computer logic includes a clock that supplies a pulse, or interrupt, every T seconds, and the A/D converter sends a number to the computer each time the interrupt arrives. An alternative implementation, often referred to as free running, is to access the A/D converter after each cycle of code execution has been completed. In the former case the sample period is precisely fixed; in the latter case the sample period is fixed essentially by the length of the code, provided that no logic branches are present, which could vary the amount of code executed.

There also may be a sampler and an A/D converter for the input command $r(t)$, which produces the discrete $r(kT)$, from which the sensed output $m(kT)$ will be subtracted to arrive at the discrete error signal $e(kT)$.

The continuous compensation is approximated by difference equations, which are the discrete version of differential equations and can be made to duplicate the dynamic behavior of the continuous compensator if

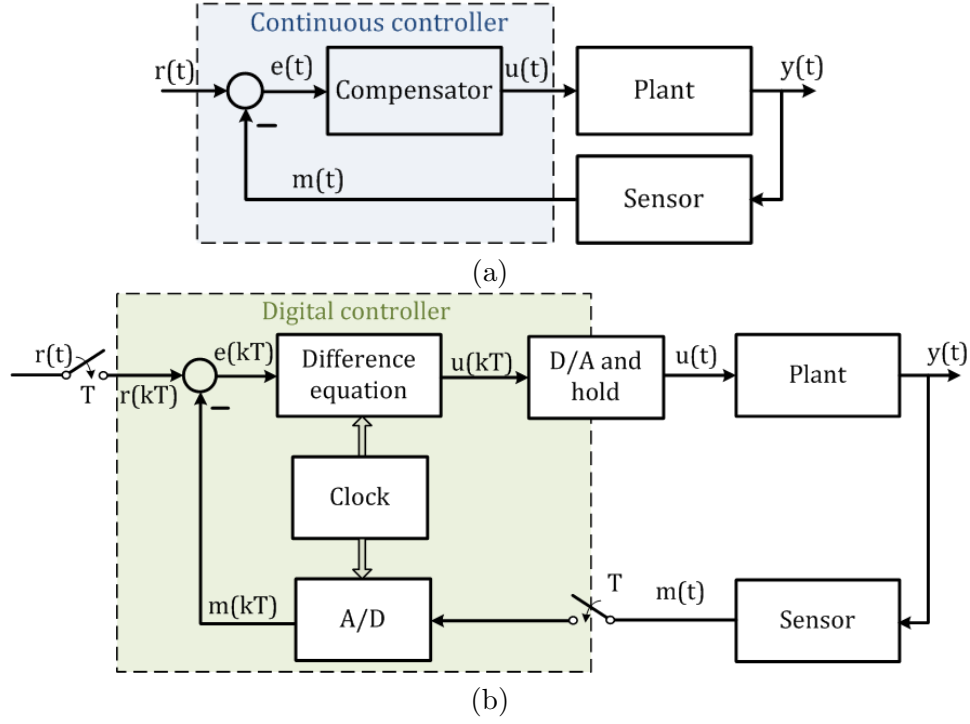


Figure 6.1: Block diagrams for a basic control system: (a) continuous and (b) with a digital computer, (Franklin et al., 2006)

the sample period is short enough. The result of the difference equations is a discrete $u(kT)$ at each sample instant. This signal is converted to a continuous $u(t)$ by the digital-to-analog (D/A) converter and the hold: the D/A converter changes the binary number to an analog voltage, and a zero-order hold maintains that same voltage throughout the sample period. The resulting $u(t)$ is then applied to the actuator in precisely the same manner as the continuous implementation, (Franklin et al., 2006).

There are two basic techniques for finding the difference equations for the digital controller:

- One technique, called discrete equivalent, consists of designing a continuous compensation using methods described in the previous chapters, then approximating the compensator transfer function using methods that will be described in the following sections.
- The other technique is discrete design. Here the difference equations are found directly without designing the continuous compensator first.

6.1.2 Sampling

In choosing the sampling period T , or the sampling frequency $\omega_s = 2\pi/T$, most authors approaching this field refer to Shannon's sampling theorem. It states that:

'A function $f(t)$ that has a bandwidth ω_b is uniquely determined by a discrete set of sample values provided that the sampling frequency is greater than $\omega_s = 2\omega_b$. The sampling frequency $\omega_s = 2\omega_b$ is called the *Nyquist frequency*' (Burns, 2001).

It is rare in practical situations to work near the limit given by Shannon's theorem. *A useful rule in applications is to sample the signal at about 10 times higher than the highest frequency thought to be present* (Burns, 2001).

Example 6.1.1 • For the original signal in Figure 6.2(a): $u(t) = \cos 2t$, the frequency is $\omega_b = 2$ rad/s.

- The sampling frequency was chosen the Nyquist frequency: $\omega_s = 2\omega_b = 4$ rad/s, that is sufficient to capture each peak (positive and negative) of the signal. The sampling period is: $T = 2\pi/\omega_s = 1.57$ s.
- If we sample the signal higher than the Nyquist frequency, with $\omega_s = 10\omega_b = 20$ rad/s, i.e the sampling period is $T = 2\pi/\omega_s = 0.314$ s, then, there are more than enough samples to capture all variations in the signal.
- For the signal shown in Figure 6.2(b): $u(t) = \cos t + \cos 2t + \cos 4t$, the highest frequency is $\omega_b = 4$ rad/s.
 - The sampling frequency was chosen the Nyquist frequency: $\omega_s = 2\omega_b = 8$ rad/s, that is sufficient to reproduce the peaks of the original signal. The sampling period is: $T = 2\pi/\omega_s = 0.78$ s.
 - If we sample the signal higher than the Nyquist frequency, with $\omega_s = 10\omega_b = 40$ rad/s, i.e the sampling period is $T = 2\pi/\omega_s = 0.157$ s, then, all variations in the signals are captured.

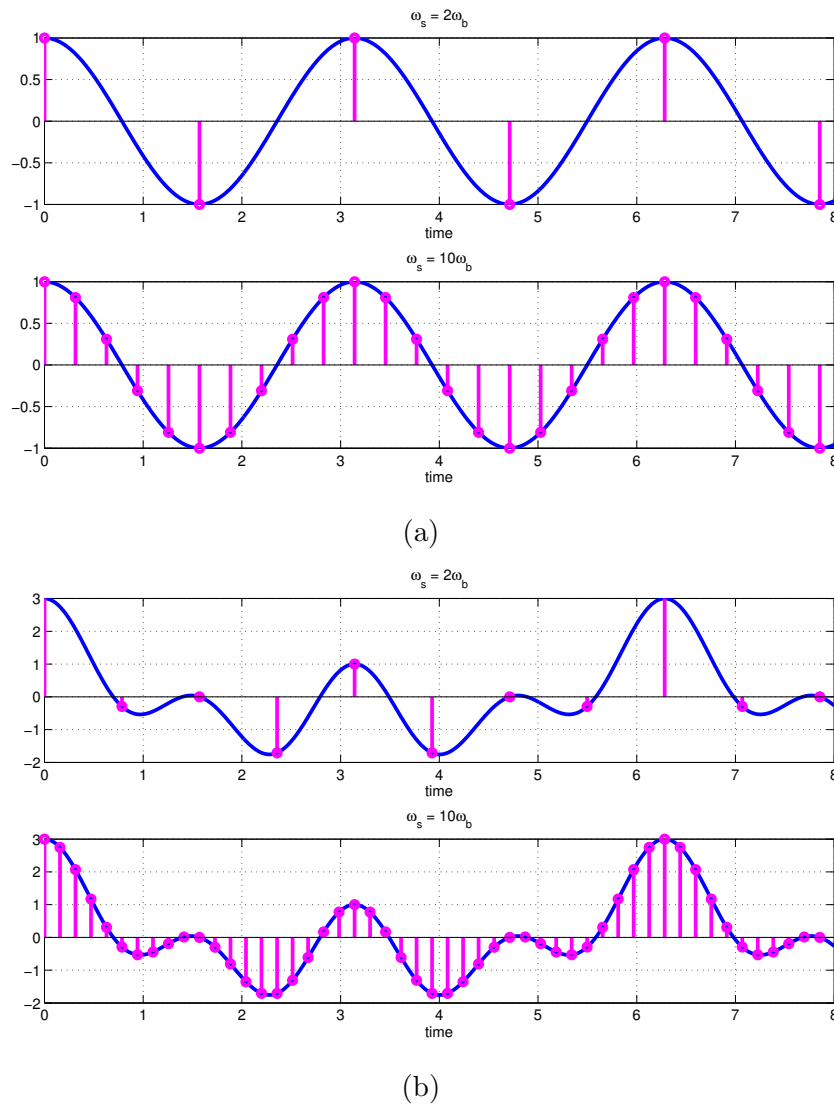


Figure 6.2: Sampled signals. (a) $u(t) = \cos 2t$, (b) $u(t) = \cos t + \cos 2t + \cos 4t$

If a signal is sampled below Shannon's limit, then a lower frequency signal, called an *alias* may be constructed as shown in Figure 6.3.

Example 6.1.2 The original signal in Figure 6.3(a) is $u(t) = \cos(2t)$, thus the frequency is $\omega_b = 2 \text{ rad/s}$, and in Figure 6.3(b) the signal is $u(t) = \cos t + \cos 2t + \cos 4t$, with the highest frequency $\omega_b = 4 \text{ rad/s}$. The sampling frequency was chosen in both cases as $\omega_s = 1.5\omega_b$. The information on the sampled signals is misleading and apparently the signal has a different form, with a lower frequency. This situation is known as aliasing.

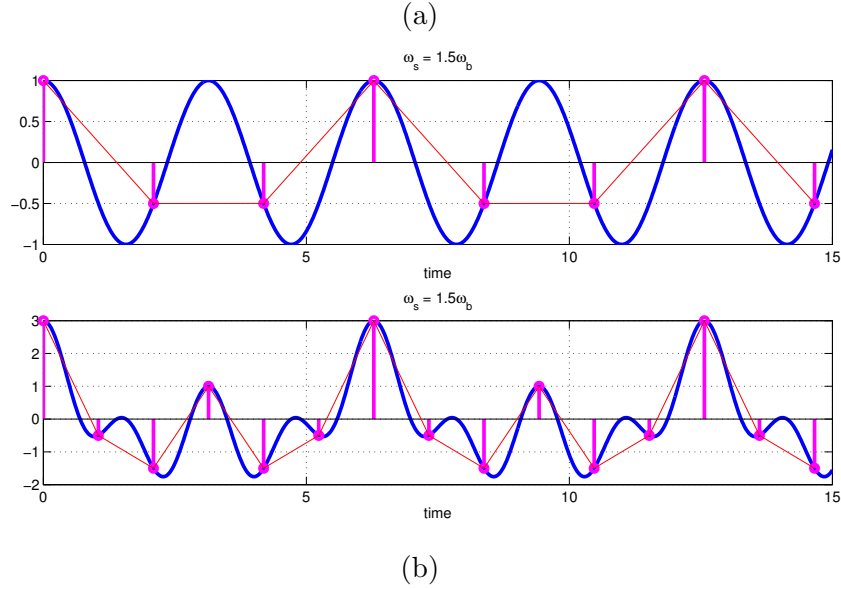


Figure 6.3: Alias due to undersampling (a) $u(t) = \cos 2t$, (b) $u(t) = \cos t + \cos 2t + \cos 4t$

To ensure that aliasing does not take place, it is common practice to place an anti-aliasing filter before the A/D converter. This is an analog low-pass filter with a break frequency of $0.5\omega_s$ (and $\omega_s > 10\omega_b$). The higher ω_s is in comparison to ω_b , the more closely the digital system resembles an analog one (Burns, 2001).

6.1.2.1 Choice of the sampling frequency for control systems

In choosing the sampling frequency (or the sampling time) for a continuous system, several approaches are available. It is rather a problem of experience than some exact procedure. In general, it has a strong influence on the dynamic properties of the controlled system as well as on the whole closed-loop system. Shannon's theorem usually gives a lower limit for the sampling frequency.

One rule is to choose the sampling frequency ω_s , as, (Landau and Zito, 2006):

$$6\omega_b < \omega_s < 25\omega_b \quad (\text{rad/s})$$

or

$$6f_b < \omega_s < 25f_b \quad (\text{Hz})$$

This corresponds to the existence of two to nine samples on the rise time of a step response, (Landau and Zito, 2006).

Although the methods to obtain a discrete equivalent of a continuous transfer function will be presented in the following sections, the choice of the sampling time for a continuous system is illustrated here with some examples.

Example 6.1.3 First-order system

$$G_1(s) = \frac{1}{T_0s + 1}$$

In this case the system bandwidth is:

$$\omega_B = 2\pi f_B = \frac{1}{T_0} \Rightarrow f_B = \frac{1}{2\pi T_0}$$

(The relationship between the bandwidth and the time constant of the first-order system is detailed in the chapter Frequency response.)

The rule for choosing the sampling frequency given above can be written in terms of time constant T_0 and sampling time T_s :

$$\frac{6}{2\pi T_0} < \frac{1}{T_s} < \frac{25}{2\pi T_0}$$

or, (with some approximation):

$$\frac{1}{T_0} < \frac{1}{T_s} < \frac{4}{T_0} \Rightarrow \frac{T_0}{4} < T_s < T_0$$

Example 6.1.4 Consider a first order and a second order system with the transfer functions:

$$G_1(s) = \frac{1}{T_0 s + 1}, \quad G_2(s) = \frac{1}{\omega_n^2 s + \frac{2\zeta}{\omega_n} s + 1}$$

For $T_0 = 1$, $\zeta = 0.7$ and $\omega_n = 1$. For these values of system parameters, the systems' bandwidth frequency may be approximated by $\omega_{b1} = 1/T_0$ and $\omega_{b2} = \omega_n$ (an attenuation higher than 3dB is introduced for frequencies higher than ω_b).

Note that for other values of the system parameters (gain, damping factor, natural frequency) the system bandwidth may be different from the approximations we made before.

The Shannon theorem will give the following sampling frequency (and the sampling period), for each system:

$$\omega_{s1} = 2\omega_{b1} = \frac{2\pi}{T_s}, \quad T_s = \frac{2\pi}{2\omega_{b1}} = \frac{2\pi T_0}{2} = \pi T_0$$

$$\omega_{s2} = 2\omega_{b2} = \frac{2\pi}{T_s}, \quad T_s = \frac{2\pi}{2\omega_{b2}} = \frac{2\pi}{2\omega_n} = \frac{\pi}{\omega_n}$$

The step responses of these two systems are presented in Figure 6.4 - continuous system response and the discrete system response.

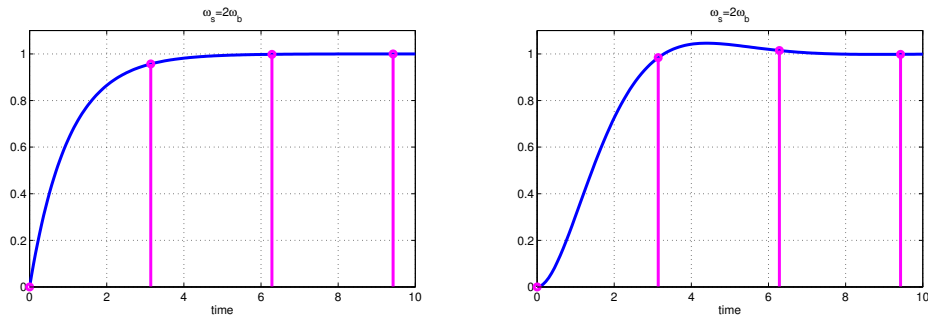


Figure 6.4: Step response of sampled systems. Left: G_1 , right: G_2

If for the first-order system the sampling frequency is chosen as $\omega_s = 6\omega_b$, the sampling period is computed as follows:

$$\omega_{s1} = \frac{2\pi}{T_s} = 6\omega_{b1} = \frac{6}{T_0}, \Rightarrow T_s = \frac{2\pi T_0}{6}$$

If the sampling frequency is chosen as $\omega_s = 25\omega_b$, then, according to the same calculation, the sampling period is $T_s = \frac{2\pi T_0}{25}$

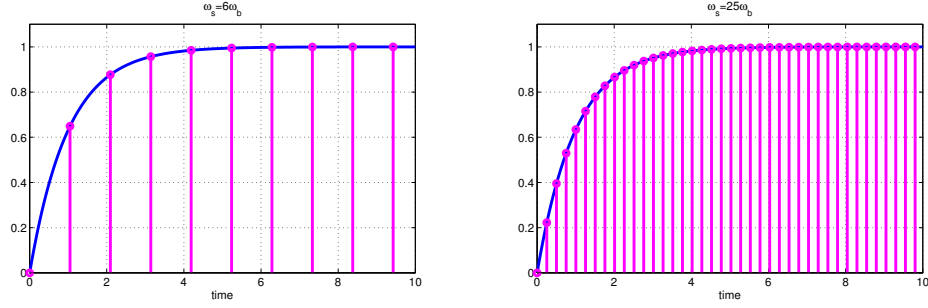


Figure 6.5: Step response of a sampled first-order system

If $T_0 = 1$, the sampling period is $T_s = 1.04$ (when $\omega_s = 6\omega_{b1}$), or $T_s = 0.25$ (when $\omega_s = 25\omega_{b1}$). The step response for each case is presented in Figure 6.5, compared to the step response of the continuous system.

If for the second-order system the sampling frequency is chosen as $\omega_s = 6\omega_b$, the sampling period is computed as follows:

$$\omega_{s2} = \frac{2\pi}{T_s} = 6\omega_{b2} = 6\omega_n, \Rightarrow T_s = \frac{2\pi}{6\omega_n}$$

If the sampling frequency is chosen as $\omega_s = 25\omega_b$, then, according to the same calculation, the sampling period is $T_s = \frac{2\pi}{25\omega_n}$

Let $\omega_n = 1$ rad/sec and $\zeta = 0.7$. Then the sampling period is $T_s = 1.04$ (when $\omega_s = 6\omega_{b2}$), or $T_s = 0.25$ (when $\omega_s = 25\omega_{b2}$). The step response for each case is presented in Figure 6.6, compared to the step response of the continuous system.

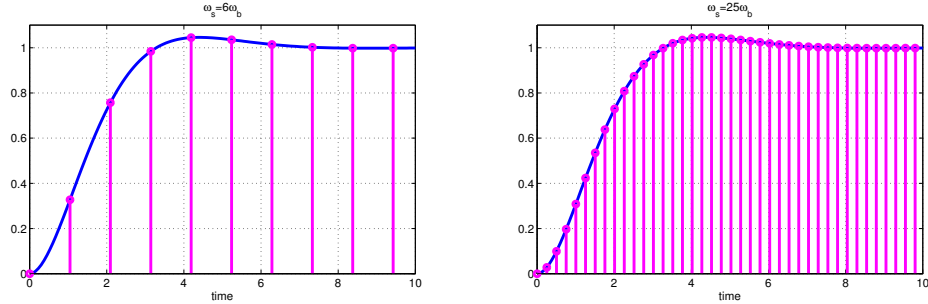


Figure 6.6: Step response of a sampled second-order system

6.1.2.2 The sampler

A **sampler** is basically a switch that closes every T seconds for one instant of time.

Consider an ideal sampler as shown in Figure 6.7.

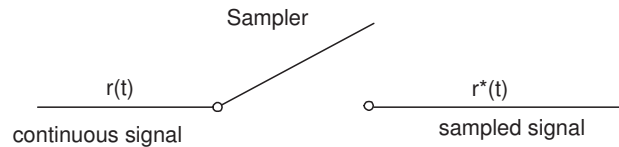


Figure 6.7: An ideal sampler with an input $r(t)$

The input is $r(t)$ and the output $r^*(t)$, where nT is the current sample time and the current value of $r^*(t)$ is $r(kT)$.

For a given time t , we have:

$$r^*(t) = r(kT) \cdot \delta(t - kT)$$

where $\delta(t - kT)$ is the unit impulse function occurring at $t = kT$.

Then we portray the series for $r^*(t)$ as a string of impulses starting at $t = 0$ spaced at T seconds and of amplitude $r^*(kT)$.

$$r^*(t) = \sum_{k=0}^{\infty} r(kT) \delta(t - kT)$$

For example see Figure 6.8.

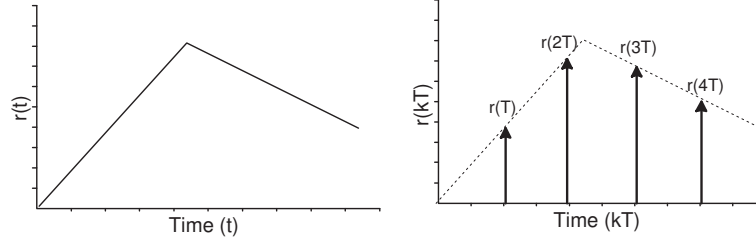


Figure 6.8: (a) An input $r(t)$ and (b) the sampled signal

6.1.3 Signal reconstruction

A digital-to-analog (D/A) converter serves as a device that converts the sampled signal $r^*(t)$ to a continuous signal $p(t)$. The D/A converter can usually be represented by a zero-order hold (ZOH) circuit, as shown in Figure 6.9.

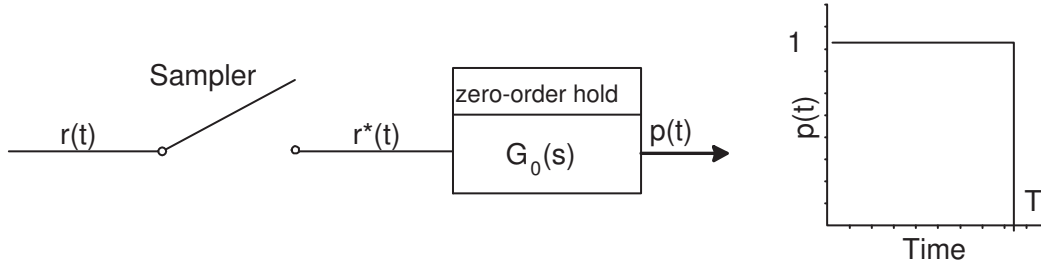


Figure 6.9: A sampler and zero-order circuit (left); the response of a zero-order hold to an impulse input $r(kT)$, which equals 1 when $k=0$ and equals zero when $k \neq 0$ (right)

The ZOH takes the value $r(kT)$ and holds it constantly for $kT \leq t \leq (k+1)T$ as shown in Figure 6.9 for $k=0$. Thus we use $r(kT)$ during the sampling period.

A sampler and ZOH can accurately follow the input signal if T is small compared to the transient changes in the signal. The response of a sampler and ZOH for a ramp input is shown in Figure 6.10.

A ZOH converts a series of impulses into a series of pulses of width T . As shown in Figure 6.11, a unit impulse, at time t , is converted into a pulse of width T , which may be created by a positive unit step at time t ($1(t)$) followed by a negative unit step ($1(t - T)$).

Then the transfer function of the zero-order hold is:

$$G_0(s) = \frac{\mathcal{L}\{u(t)\}}{\mathcal{L}\{\delta(t)\}} = \mathcal{L}\{1(t) - 1(t - T)\} = \frac{1}{s} - \frac{1}{s}e^{-sT} = \frac{1 - e^{-sT}}{s} \quad (6.1)$$

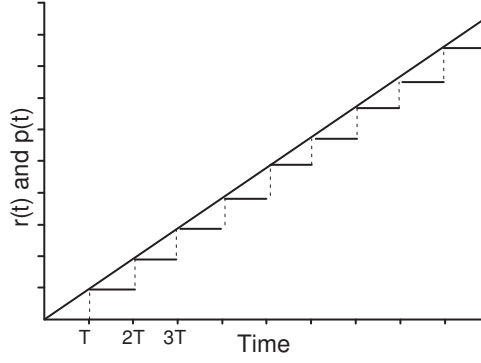


Figure 6.10: The response of a sampler and zero-order hold for a ramp input $r(t)=t$

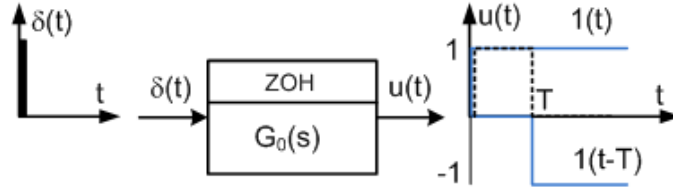


Figure 6.11: The impulse response of a ZOH

6.1.4 The z-Transform

The z-transform is the main analytical tool for SISO discrete-time systems and it analogous to the Laplace transform for continuous systems. The symbol z can be associated with a discrete time shifting in a difference equation in the same way that s can be associated with differentiation in a differential equation.

Because the output of the ideal sampler, $r^*(t)$, is a series of impulses with values $r(kT)$ we have:

$$r^*(t) = \sum_{k=0}^{\infty} r(kT) \delta(t - kT) \quad (6.2)$$

for a signal for $t > 0$. Using the Laplace transform, we have:

$$\mathcal{L}[r^*(t)] = \sum_{k=0}^{\infty} r(kT) e^{-ksT} \quad (6.3)$$

We now have an infinite series that involves factors of e^{sT} and its powers. We define

$$z = e^{sT} \quad (6.4)$$

where this relationship involves a conformal mapping from the s-plane to the z-plane. We then define a new transform, called the **z-Transform**, so that:

$$Z[r(t)] = Z[r^*(t)] = R(z) = \sum_{k=0}^{\infty} r(kT) z^{-k} \quad (6.5)$$

As an example, let us determine the z-transform of the unit step function $u(t)=1$. We obtain:

$$Z[u(t)] = \sum_{k=0}^{\infty} u(kT) z^{-k} = \sum_{k=0}^{\infty} z^{-k} \quad (6.6)$$

since $u(kT)=1$, for $k \geq 0$. This series can be written in closed form as:

$$U(z) = \frac{1}{1 - z^{-1}} = \frac{z}{z - 1} \quad (6.7)$$

Obs. Recall that the infinite geometric series may be written: $(1 - bx)^{-1} = 1 + bx + (bx)^2 + \dots$ if $(bx)^2 < 1$

Example 6.1.5 Transform of exponential

Let us determine the z -transform of $f(t) = e^{-at}$, $t \geq 0$

$$Z[e^{-at}] = F(z) = \sum_{k=0}^{\infty} e^{-akT} z^{-k} = \sum_{k=0}^{\infty} (ze^{aT})^{-k} \quad (6.8)$$

Again, this series can be written in closed form as:

$$F(z) = \frac{1}{1 - (ze^{aT})^{-1}} = \frac{z}{z - e^{-aT}} \quad (6.9)$$

In general we may show that

$$Z[e^{-at} f(t)] = F(e^{aT} z) \quad (6.10)$$

Table 6.1: z -Transforms

$x(t)$	$X(s)$	$X(z)$
$\delta(t) = \begin{cases} 1, t = 0 \\ 0, t = kT, k \neq 0 \end{cases}$	1	1
$\delta(t - kT) = \begin{cases} 1, t = kT \\ 0, t \neq kT, \end{cases}$	e^{-kTs}	z^{-k}
$u(t)=1$, unit step	$\frac{1}{s}$	$\frac{z}{z-1}$
t	$\frac{1}{s^2}$	$\frac{Tz}{(z-1)^2}$
e^{-at}	$\frac{1}{s+a}$	$\frac{z}{z-e^{-aT}}$
$1 - e^{-at}$	$\frac{1}{s(s+a)}$	$\frac{(1-e^{-aT})z}{(z-1)(z-e^{-aT})}$
$\sin \omega t$	$\frac{\omega}{s^2 + \omega^2}$	$\frac{z \sin \omega T}{z^2 - 2z \cos \omega T + 1}$

6.1.5 The pulse transfer function

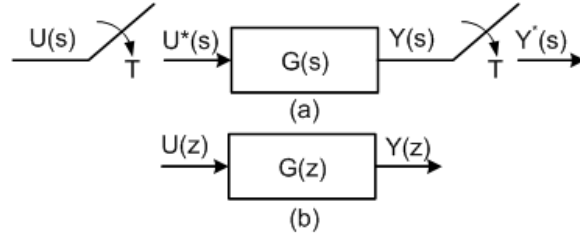
Consider the block diagram shown in Figure 6.12. The signal $U^*(s)$ is a sampled input to $G(s)$ which gives a continuous output $Y(s)$ which when sampled becomes $Y^*(s)$. Figure (b) shows the pulse transfer function where $U(z)$ is equivalent to $U^*(s)$ and $Y(z)$ is equivalent to $Y^*(s)$, (Burns, 2001).

The *pulse transfer function* is

$$G(z) = \frac{Y(z)}{U(z)} \quad (6.11)$$

Table 6.2: Properties of z-transform

$x(t)$	$X(z)$
$kx(t)$	$kX(z)$
$x_1(t) + x_2(t)$	$X_1(z) + X_2(z)$
$x(t + T)$	$zX(z) - zx(0)$
$tx(t)$	$-Tz \frac{dX(z)}{dz}$
$e^{-at}x(t)$	$X(ze^{aT})$
$x(0)$, initial value	$\lim_{z \rightarrow \infty} X(z)$ if the limit exists
$x(\infty)$, final value	$\lim_{z \rightarrow 1} (z - 1)X(z)$ if the limit exists and the system is stable

Figure 6.12: $G(s)$ and $G(z)$

6.1.6 Blocks in cascade

In Figure 6.13 there are samplers either side of the blocks $G_1(s)$ and $G_2(s)$. The pulse transfer function is:

$$\frac{Y(z)}{U(z)} = G_1(z)G_2(z) \quad (6.12)$$

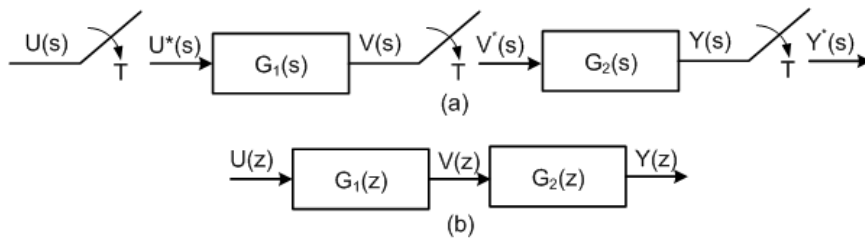


Figure 6.13: Blocks in cascade - all signals sampled

In figure 6.14 there is no sampler between $G_1(s)$ and $G_2(s)$ so they can be combined to give $G_1(s)G_2(s) = G_1G_2(s)$. The pulse transfer function is computed as:

$$\frac{Y(z)}{U(z)} = G_1G_2(z) = Z\{G_1G_2(s)\} \quad (6.13)$$

Note that $G_1(z)G_2(z) \neq G_1G_2(z)$.

6.1.7 Inverse transformation

The discrete time response can be found using a number of methods.

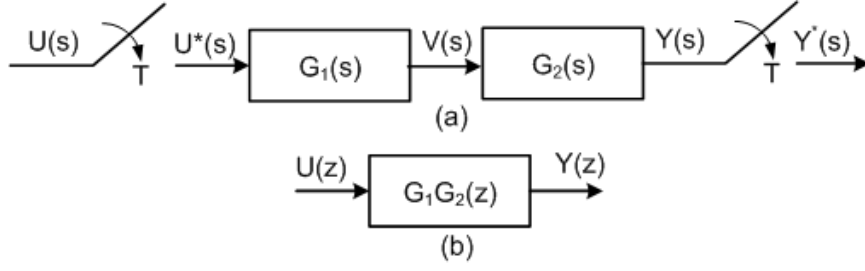


Figure 6.14: Blocks in cascade - input and output sampled

6.1.7.1 Infinite power series method

Example 6.1.6 If the z -transform of a signal is obtained in the form:

$$X(z) = \frac{z^2}{z^2 - 1.36z + 0.36}, \quad (6.14)$$

by long division we obtain:

$$X(z) = z^2 : (z^2 - 1.36z + 0.36) = 1 + 1.36z^{-1} + 1.5z^{-2} + \dots \quad (6.15)$$

From equation (6.5) we have:

$$X(z) = \sum_{k=0}^{\infty} x(kT)z^{-k} = x(0)z^0 + x(T)z^{-1} + x(2T)z^{-2} + \dots \quad (6.16)$$

Thus:

$$x(0) = 1, \quad x(T) = 1.36, \quad x(2T) = 1.5, \quad \text{etc} \quad (6.17)$$

6.1.7.2 Difference equation method

Consider a system of the form:

$$G(z) = \frac{Y(z)}{X(z)} = \frac{b_0 + b_1z^{-1} + b_2z^{-2} + \dots}{1 + a_1z^{-1} + a_2z^{-2} + \dots} \quad (6.18)$$

By cross-multiplication we get:

$$(1 + a_1z^{-1} + a_2z^{-2} + \dots)Y(z) = (b_0 + b_1z^{-1} + b_2z^{-2} + \dots)X(z) \quad (6.19)$$

or

$$Y(z) = -(a_1z^{-1} + a_2z^{-2} + \dots)Y(z) + (b_0 + b_1z^{-1} + b_2z^{-2} + \dots)X(z) \quad (6.20)$$

$$Y(z) = -a_1z^{-1}Y(z) + a_2z^{-2}Y(z) + \dots + b_0X(z) + b_1z^{-1}X(z) + b_2z^{-2}X(z) + \dots \quad (6.21)$$

The equation above can be expressed as a difference equation of the form:

$$y(kT) = -a_1y((k-1)T) + a_2y((k-2)T) + \dots + b_0x(kT) + b_1x((k-1)T) + b_2x((k-2)T) + \dots \quad (6.22)$$

or simply:

$$y_k = -a_1y_{k-1} + a_2y_{k-2} + \dots + b_0x_k + b_1x_{k-1} + b_2x_{k-2} + \dots \quad (6.23)$$

Example 6.1.7 (Burns, 2001). Consider a transfer function:

$$G(z) = \frac{Y(z)}{X(z)} = \frac{z}{z - 0.368} \quad (6.24)$$

The equation (6.24) can be written as:

$$\frac{Y(z)}{X(z)} = \frac{1}{1 - 0.368z^{-1}} \quad (6.25)$$

or

$$(1 - 0.368z^{-1})Y(z) = X(z) \quad \text{or} \quad Y(z) = 0.368z^{-1}Y(z) + X(z) \quad (6.26)$$

It can be expressed as a difference equation:

$$y_k = 0.368y_{k-1} + x_k \quad (6.27)$$

If the first two samples are known, the values for y_k can be determined by an iterative procedure.

6.1.8 Digitizing analog transfer functions

6.1.8.1 Zero-order hold - ZOH

The conversion of an analog transfer function to a digital one is generally best done by one of the approximations of the z-transform. Just as the Laplace transform maps continuous time quantities into the complex frequency domain, thus allowing linear transfer function models to be written in terms of the variable s , so the z-transform maps linear transfer function models into discrete time, and allows transfer function models for discrete time systems to be written in terms of a variable z .

One approach in digitizing an analog transfer function is transformation into z-domain, as described in the previous subsections. Given an s-domain transfer function $G(s)$, the z-transform can be obtained from:

$$G(z) = Z(G_0(s)G(s)) = Z\left(\frac{1 - e^{-sT}}{s}G(s)\right) = (1 - z^{-1})Z\left(\frac{G(s)}{s}\right) \quad (6.28)$$

where $G_0(s)$ is the transfer function of a zero-order hold, and T is the sampling time.

Example 6.1.8 Let us consider the system in Figure 6.15, for $T = 1$.

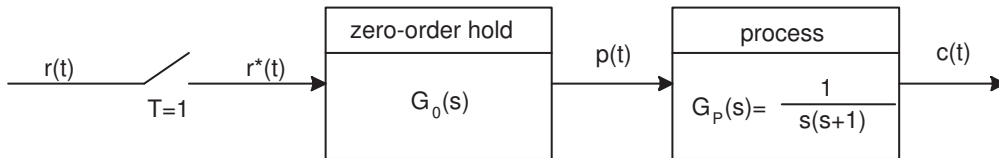


Figure 6.15: An open-loop sampled data system

The transfer function of a ZOH is

$$G_0(s) = \frac{1 - e^{-sT}}{s}$$

Therefore the transfer function $C(s)/R^*(s)$ is

$$\frac{C(s)}{R^*(s)} = G_0(s)G_P(s) = G(s) = \frac{1 - e^{-sT}}{s^2(s+1)} \quad (6.29)$$

Expanding into partial fractions, we have:

$$G(s) = (1 - e^{-sT}) \left(\frac{1}{s^2} - \frac{1}{s} + \frac{1}{s+1} \right) \quad (6.30)$$

$$G(z) = Z[G(s)] = (1 - z^{-1})Z \left[\frac{1}{s^2} - \frac{1}{s} + \frac{1}{s+1} \right] \quad (6.31)$$

Using the entries of Table 1 to obtain the z-transform of each term we have:

$$G(z) = Z[G(s)] = (1 - z^{-1}) \left(\frac{Tz}{(z-1)^2} - \frac{z}{z-1} + \frac{z}{z-e^{-aT}} \right) = \frac{(ze^{-T} - z + Tz) + (1 - e^{-T} - Te^{-T})}{(z-1)(z-e^{-T})} \quad (6.32)$$

When $T=1$, we obtain:

$$G(z) = \frac{ze^{-1} + 1 - 2e^{-1}}{(z-1)(z-e^{-1})} = \frac{0.3678z + 0.2644}{(z-1)(z-0.3678)} = \frac{0.3678z + 0.2644}{z^2 - 1.3678z + 0.3678} \quad (6.33)$$

The response of this system to unit impulse is obtained for $R(z)=1$ so that $C(z)=G(z) \cdot 1$. We may obtain $C(z)$ by dividing the denominator into the numerator as:

$$C(z) = 0.3678z + 0.2644 : z^2 - 1.3678z + 0.3678 = 0.3678z^{-1} + 0.7675z^{-2} + 0.9145z^{-3} + \dots \quad (6.34)$$

The calculation yields the response at the sampling instants and can be carried out as far as is needed for $C(z)$. From equation (6.5) we have:

$$C(z) = \sum_{k=0}^{\infty} c(kT)z^{-k} \quad (6.35)$$

In this case we have obtained $c(kT)$ as follows: $c(0) = 0$, $c(T) = 0.3678$, $c(2T) = 0.7675$, $c(3T) = 0.9145 \dots$

Note that $c(kT)$ provides the values of $c(t)$ at $t = kT$.

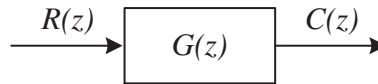


Figure 6.16: The z-transform transfer function in block diagram

We have determined $C(z)$, the z-transform of the output sampled signal. The z-transform of the input signal is $R(z)$. The transfer function in the z-domain is:

$$\frac{C(z)}{R(z)} = G(z) \quad (6.36)$$

6.1.8.2 Simple substitution

This variable z , strictly arises from a discrete time summation expression related to the definition of the Laplace transform. However, it is usually approximated by various substitutions for s in the linear transfer function model. These different possible derivations lead to different possible z-transform models of the same system, each having slightly different properties.

The variable z is first introduced in the simplest manner possible. That is, a multiplication by z^{-1} is used as a notation to represent a *delay* of one time step in a discrete-time signal.

In obtaining z-transfer functions from linear transfer functions by making substitutions for s , many of the generally used substitutions depend on the property that multiplying by s implies differentiation in

the time domain. Multiplying the z-transform by z^{-1} implies an additional delay of one sampling interval. The argument for the simplest of the substitutions suggests that the gradient of the time response of, for example, a signal $e(t)$ at the present time (instant k) is given by:

$$\left. \frac{de}{dt} \right|_k = \frac{(\text{latest_sample_of_}e) - (\text{last_sample_of_}e)}{\text{sampling_interval}} = \frac{e_k - e_{k-1}}{T} \quad (6.37)$$

In the z-domain, when e_k becomes $E(z)$ (that is, the z-transform of e_k) this will convert to:

$$\frac{E(z) - E(z) \cdot z^{-1}}{T}$$

or

$$E(z) \frac{1 - z^{-1}}{T}$$

Multiplying $E(z)$ by $\frac{1-z^{-1}}{T}$ is therefore equivalent to differentiation, that is, to multiplying $E(s)$ by s . This means that substituting the following expressions for s in a linear transfer function:

$$s = \frac{1 - z^{-1}}{T} \quad (6.38)$$

will give an approximately equivalent discrete-time version (in z) for the analog linear transfer function. It is then easy to convert the z-transfer function to an algorithm from which a computer program can be written.

6.1.8.3 Tustin substitution

A more exact substitution for s (known as the Tustin transformation) is:

$$s = \frac{2(1 - z^{-1})}{T(1 + z^{-1})} \quad (6.39)$$

This more complicated substitution often allows close approximation of the behavior of the analog controller using a longer sampling interval (T) than would be the case with the simpler substitution for s . The longer sampling interval is beneficial in that either a slower processor can be used to perform the calculation, or a given processor can cope with a faster sampling rate.

Example 6.1.9 Consider a continuous system having the transfer function:

$$G(s) = \frac{1}{s + 4} \quad (6.40)$$

First we shall determine the pulse transfer function using the ZOH method:

$$G_1(z) = Z\{G_0(s)G(s)\} = Z\left\{\frac{1 - e^{-sT}}{s} \frac{1}{s + 4}\right\} = (1 - z^{-1}) \frac{1}{4} Z\left\{\frac{1}{s} - \frac{1}{s + 4}\right\} \quad (6.41)$$

Using Table 1 to determine the z-transforms of basic signals we obtain:

$$G_1(z) = \frac{z - 1}{z} \frac{1}{4} \left(\frac{z}{z - 1} - \frac{z}{z - e^{-4T}} \right) = \frac{1}{4} \frac{1 - e^{-4T}}{z - e^{-4T}} \quad (6.42)$$

Using the simple substitution $s = \frac{1-z^{-1}}{T}$, the pulse transfer function is:

$$G_2(z) = \frac{1}{\frac{1-z^{-1}}{T} + 4} = \frac{T}{1 + 4T - z^{-1}} \quad (6.43)$$

Using the Tustin substitution, the pulse transfer function result as:

$$G_3(z) = \frac{1}{\frac{2(1-z^{-1})}{T(1+z^{-1})} + 4} = \frac{T(1+z^{-1})}{2+4T+(4T-2)z^{-1}} \quad (6.44)$$

For $T = 0.1$, the pulse transfer functions (written in z instead of z^{-1}) in all cases above are:

$$G_1(z) = \frac{0.08242}{z - 0.6703}, \quad G_2(z) = \frac{0.1z}{1.4z - 1}, \quad G_3(z) = \frac{0.04167z + 0.04167}{z - 0.6667} \quad (6.45)$$

The simulated unit step response in all three cases is shown in Figure 6.17.

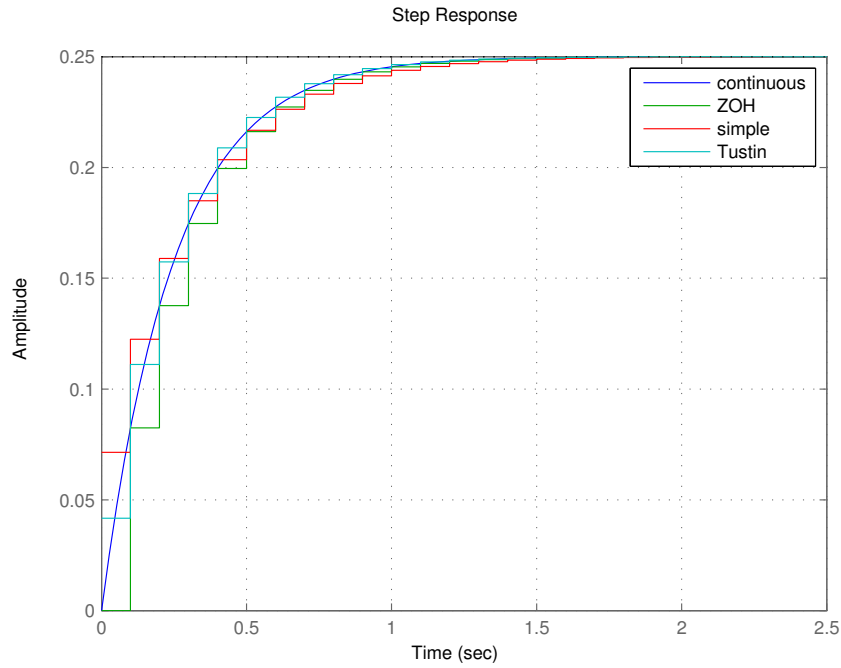


Figure 6.17: Step response

Example 6.1.10 Systems in cascade and the pulse transfer function

Determine the overall pulse transfer function for the systems shown in Figure 6.18. Compare cases (a) and (b).

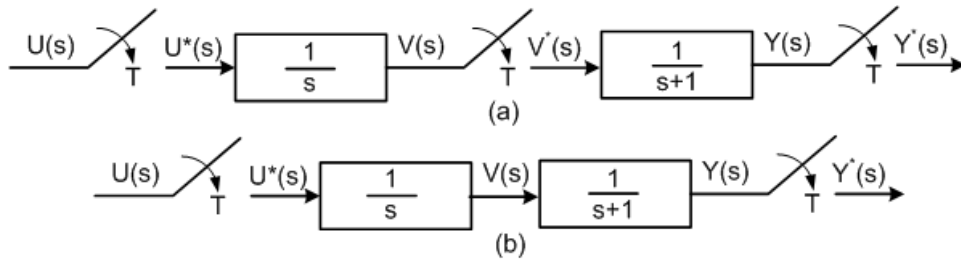


Figure 6.18: Systems in cascade

Solution. In case (a), all signals are sampled, therefore we can determine a pulse transfer function for each of the blocks shown in Figure 6.18. For case (b), an overall transfer function for the continuous systems

must be computed and the pulse transfer function will be determined as explained in Chapter 6 (section 4).
Let

$$G_1(s) = \frac{1}{s}, \text{ and } G_2(s) = \frac{1}{s+1}$$

(a) In the first case:

$$G_1(z) = (1 - z^{-1})Z\left(\frac{G_1(s)}{s}\right) = (1 - z^{-1})Z\left(\frac{1}{s^2}\right) = (1 - z^{-1})\frac{Tz}{(z-1)^2} = \frac{T}{z-1}$$

$$G_2(z) = (1 - z^{-1})Z\left(\frac{G_2(s)}{s}\right) = (1 - z^{-1})Z\left(\frac{1}{s(s+1)}\right) = (1 - z^{-1})Z\left(\frac{1}{s} - \frac{1}{s+1}\right)$$

or

$$G_2(z) = \frac{z-1}{z} \left(\frac{z}{z-1} - \frac{z}{z-e^{-T}} \right) = \frac{1-e^{-T}}{z-e^{-T}}$$

The overall transfer function is

$$G_{0a}(z) = G_1(z)G_2(z) = \frac{T}{z-1} \frac{1-e^{-T}}{z-e^{-T}}$$

(b) In the second case, the continuous equivalent transfer function is:

$$G_1G_2(s) = \frac{1}{s(s+1)}$$

Then:

$$G_{0b}(z) = (1 - z^{-1})Z\left\{\frac{G_1G_2(s)}{s}\right\} = (1 - z^{-1})Z\left\{\frac{1}{s^2(s+1)}\right\} = (1 - z^{-1})Z\left\{\frac{1}{s+1} + \frac{1}{s^2} - \frac{1}{s}\right\}$$

$$G_{0b}(z) = \frac{z-1}{z} \left(\frac{z}{z-e^{-T}} + \frac{Tz}{(z-1)^2} - \frac{z}{z-1} \right)$$

and:

$$G_{0b}(z) = \frac{z-1}{z-e^{-T}} + \frac{T}{z-1} - 1 = \frac{(z-1)^2 + (z-e^{-T}) - (z-1)(z-e^{-T})}{(z-1)(z-e^{-T})} = \frac{ze^{-T} + 1 - 2e^{-T}}{(z-1)(z-e^{-T})}$$

It is clear that $G_{0a}(z) \neq G_{0b}(z)$ for any positive value of T .

6.1.9 Closed-Loop Sampled Data Systems

Consider the sampled data z-transform model of a system with sampled output signal $C(z)$ shown in Figure 6.19. The closed-loop transfer function $T(z)$ (using block diagram reduction) is:

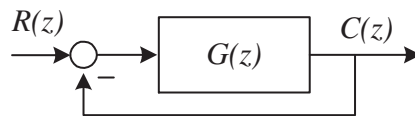


Figure 6.19: Feedback control system with unity feedback

$$\frac{C(z)}{R(z)} = T(z) = \frac{G(z)}{1 + G(z)} \quad (6.46)$$

We assume that $G(z)$ is the z-transform of $G_0(s)G_p(s)$, where $G_0(s)$ is the zero-order hold and $G_p(s)$ is the plant transfer function.

A feedback control system with a digital controller is shown in Figure 6.20. The z-transform of the block diagram model is:

$$\frac{C(z)}{R(z)} = T(z) = \frac{G(z)D(z)}{1 + G(z)D(z)} \quad (6.47)$$

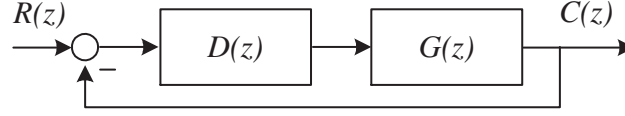


Figure 6.20: Feedback control system with digital controller

Example 6.1.11 Response of a closed-loop system

We consider a closed-loop sampled-data control system shown in Figure 6.21.

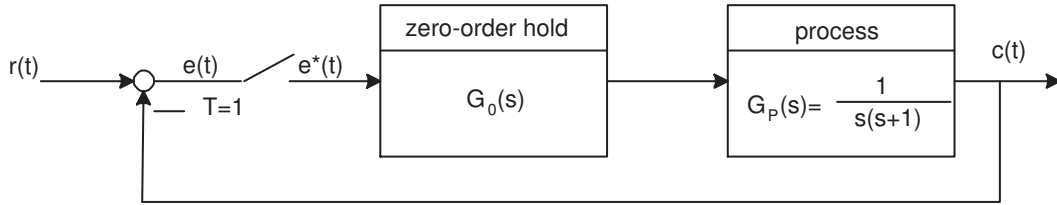


Figure 6.21: A closed-loop sampled data system

The z-transform of the open-loop system $G(z)$ was obtained in the previous Example (equation (6.33)). Therefore we have:

$$\frac{C(z)}{R(z)} = \frac{G(z)}{1 + G(z)} = \frac{0.3678z + 0.2644}{z^2 - z + 0.6322} \quad (6.48)$$

Since the input is a unit step

$$R(z) = \frac{z}{z - 1} \quad (6.49)$$

then

$$C(z) = \frac{z(0.3678z + 0.2644)}{(z - 1)(z^2 - z + 0.6322)} = \frac{0.3678z^2 + 0.2644z}{z^3 - 2z^2 + 1.6322z - 0.6322} \quad (6.50)$$

Completing the division we have:

$$C(z) = 0.3678z^{-1} + z^{-2} + 1.4z^{-3} + 1.4z^{-4} + 1.147z^{-5} \quad (6.51)$$

The values of $c(kT)$ are shown in Figure 6.22 (b). The complete response of the sampled-data closed-loop system is shown and contrasted to the response of a continuous system (where $T=0$). The overshoot of the sampled data system is 40% in contrast to 17% for a continuous system. Furthermore, the settling time is twice as long as of the continuous system.

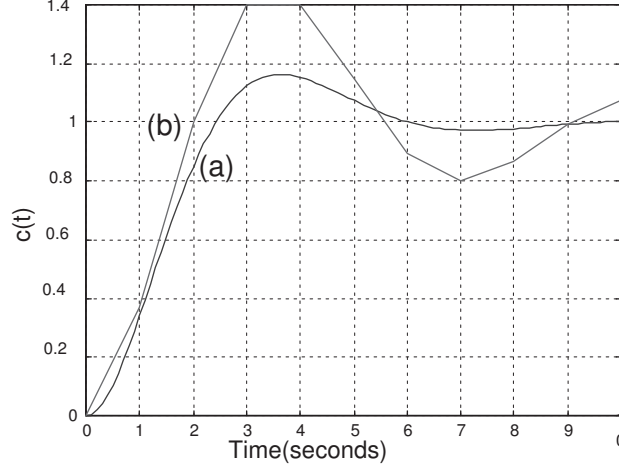


Figure 6.22: The response of a second-order system (a)continuous $T=0$ (b) sampled system $T=1$

6.2 Mapping between the s-plane and the z-plane

The s-plane and the z-plane are related by a *conformal mapping* specified by the analytic complex function

$$z = e^{sT} = e^{(\sigma + j\omega)T} = e^{\sigma T} \cdot e^{j\omega T}$$

where T is the sampling time and:

$$\text{Re}[s] = \sigma, \quad \text{Im}[s] = j\omega, \quad \text{and} \quad |z| = e^{\sigma T}, \quad \angle z = \omega T$$

- The point $s = 0$ is mapped in the z-plane to $z = e^0 = 1$
- A complex imaginary number $s = +j\omega$ is mapped into $z = e^{j\omega T}$, that is a unit length vector at angle ωT (see Figures 6.23 and 6.24). When $\omega T = \pi$, the vector reached the point -1 in z-plane, thus describing half of a circle. When $\omega T = -\pi$, the other half of the circle is obtained.
- All complex numbers $s = \sigma + j\omega$ with the real part σ constant are mapped into z-plane in circles having the radius $e^{\sigma T}$ (see Figure 6.23)

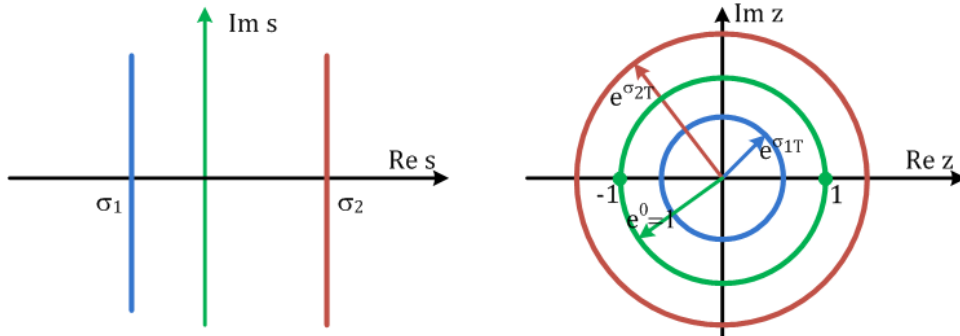


Figure 6.23: Mapping vertical lines

- Complex numbers $s = \sigma + j\omega$ with the same imaginary part $j\omega$, are mapped to radial lines with constant angle ωT (see Figure 6.24).

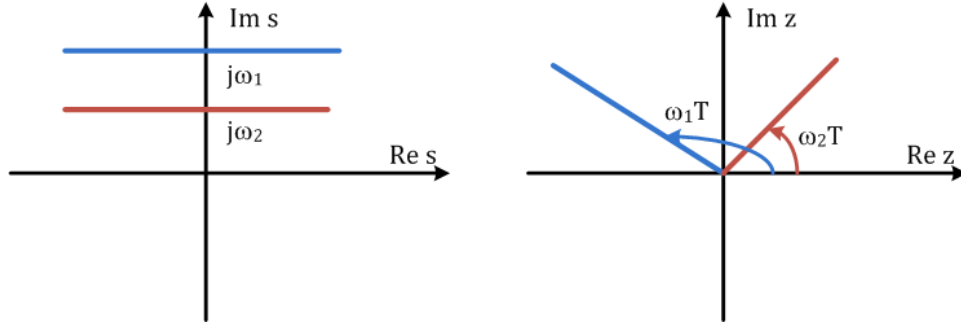


Figure 6.24: Mapping horizontal lines

6.3 Stability analysis in z-plane

A linear continuous feedback control system is stable if all the poles of the closed-loop transfer function $T(s)$ lie in the left half of the s-plane. The s-plane is related to the z-plane by the transformation:

$$z = e^{sT} = e^{(\sigma + j\omega)T} \quad (6.52)$$

We may also write this relationship as:

$$|z| = e^{\sigma T} \quad (6.53)$$

and

$$\angle z = \omega T \quad (6.54)$$

In the left-hand s-plane, the real part of s , $\sigma < 0$ and therefore the related magnitude of z varies between 0 and 1 ($0 < e^{\sigma T} < 1$). Therefore the imaginary axis of the s-plane corresponds to the unit circle in the z-plane, and the inside of the unit circle corresponds to the left half of the s-plane.

Therefore we can state that *a sampled data system is stable if all the poles of the closed-loop transfer function $T(z)$ lie within the unit circle of the z-plane.*

As shown in Figure 6.25 (Nise, 2004), each region of the s-plane can be mapped into a corresponding region on the z-plane. Points that have negative values of σ (left half s-plane, region A) map into the inside of the unit circle on the z-plane. Points on the $j\omega$ axis, region B, have zero values of σ and yield points on the unit circle on the z-plane. The points that have positive values of σ are in the right half of the s-plane, region C. The magnitudes of the mapped points are $e^{\sigma T} > 1$, thus they are mapped into points outside the circle on the z-plane.

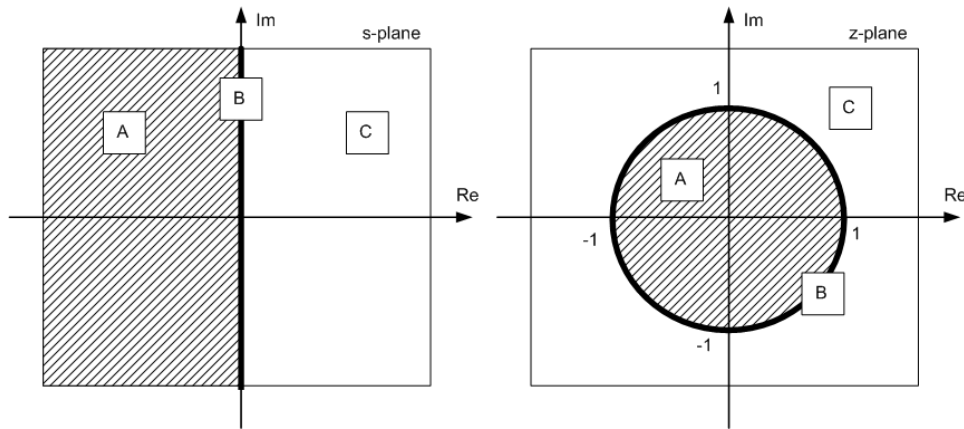


Figure 6.25: Mapping regions of the s-plane onto the z-plane

Example 6.3.1 Consider a feedback control system as the one shown in Figure 6.26, where $G(z)$ is the z -transform of open-loop transfer function:

$$G(s) = \frac{k}{s(s+1)} \quad (6.55)$$

$$G(z) = Z(G_0(s)G(s)) = Z\left(\frac{1-e^{-sT}}{s} \frac{k}{s(s+1)}\right) = (1-z^{-1})Z\left(\frac{k}{s^2(s+1)}\right) \quad (6.56)$$

For $T = 1$, we have:

$$G(z) = \frac{k(0.367z + 0.264)}{z^2 - 1.367z + 0.367} \quad (6.57)$$

The closed-loop transfer function (Figure 6.26) is calculated as:

$$T(z) = \frac{G(z)}{1 + G(z)} = \frac{k(0.367z + 0.264)}{z^2 + (0.367k - 1.367)z + 0.264k + 0.367} \quad (6.58)$$

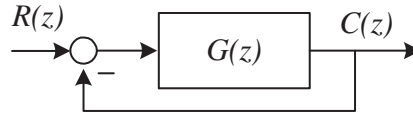


Figure 6.26: Feedback control system with unity feedback

The poles of the closed-loop transfer function $T(z)$ are the roots of the equation $q(z) = 1 + G(z) = 0$ (the characteristic equation).

When $k = 1$, we have:

$$q(z) = z^2 + (0.367k - 1.367)z + 0.264k + 0.367 = z^2 - z + 0.631 \quad (6.59)$$

The roots of $q(z) = 0$ are: $z_1 = 0.5 + 0.6173j$ and $z_2 = 0.5 - 0.6173j$. The system is stable because the roots lie within the unit circle.

When $k = 10$ we have:

$$q(z) = z^2 + (0.367k - 1.367)z + 0.264k + 0.367 = z^2 + 2.31z + 3.01 \quad (6.60)$$

The roots of $q(z) = 0$ are: $z_1 = -1.1550 + 1.2946j$ and $z_2 = -1.1550 - 1.2946j$. The system is not stable because the roots lie outside the unit circle.

6.4 Digital controllers

The general form of the pulse transfer function between an input $X(z)$ and an output $Y(z)$ is given by:

$$\frac{Y(z)}{X(z)} = \frac{b_0 + b_1 z^{-1} + \dots + b_m z^{-m}}{1 + a_1 z^{-1} + \dots + a_n z^{-n}}$$

where a_i 's and b_i 's are real coefficients. By cross multiplication we obtain:

$$Y(z)(1 + a_1 z^{-1} + \dots + a_n z^{-n}) = X(z)(b_0 + b_1 z^{-1} + \dots + b_m z^{-m})$$

In terms of difference equation:

$$y(kT) = a_1 y((k-1)T) + \dots + a_n y((k-n)T) = b_0 x(kT) + b_1 x((k-1)T) + \dots + b_m x((k-m)T)$$

or, by simplifying the notation $kT \rightarrow k$, and in general $(k-p)T \rightarrow k-p$, we obtain:

$$y(k) = a_1 y(k-1) + \dots + a_n y(k-n) = b_0 x(k) + b_1 x(k-1) + \dots + b_m x(k-m)$$

6.4.1 Digital PID from the time domain

Consider an ideal PID controller in continuous time domain, where $u(t)$ is the control signal and $e(t)$, the error signal:

$$u(t) = K_P e(t) + K_I \int_0^t e(\tau) d\tau + K_D \frac{de(t)}{dt}$$

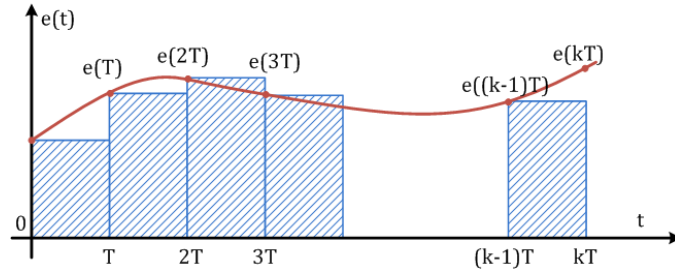


Figure 6.27: Discrete error signal

To obtain a discrete representation of the controller we need to approximate the integral and derivative terms to forms suitable for computation by a computer. With the sampling time denoted by T , the following approximations may be used (see Figure 6.27):

$$\left. \frac{de(t)}{dt} \right|_{t=kT} \approx \frac{e(kT) - e((k-1)T)}{T}$$

$$\int_0^t e(\tau) d\tau \Big|_{t=kT} \approx \sum_{n=0}^{k-1} T \cdot e(nT)$$

If the notation is simplified by taking $kT \rightarrow k$, and $(k-1)T \rightarrow k-1$ the discrete PID algorithm is then:

$$u(k) = K_P e(k) + K_I T \sum_{n=0}^{k-1} e(n) + K_D \frac{e(k) - e(k-1)}{T} \quad (6.61)$$

The difference equation (6.61) is known as *positional PID* controller.

6.4.2 Digital PID from the s-domain

The digital PID controller can also be formulated from the Laplace domain. The ideal PID algorithm is written as:

$$\frac{U(s)}{E(s)} = K_P + \frac{K_I}{s} + K_D s$$

We can apply $s \rightarrow z$ transformations to get an equivalent discrete PID controller. If, for example, the transformation is:

$$s = \frac{1 - z^{-1}}{T}$$

then, the discrete PID pulse transfer function is:

$$\frac{U(z)}{E(z)} = K_P + \frac{K_I T}{1 - z^{-1}} + K_D \frac{1 - z^{-1}}{T}$$

or

$$\frac{U(z)}{E(z)} = K_P(1 - z^{-1}) + K_I T + \frac{K_D}{T}(1 - z^{-1})^2$$

$$(1 - z^{-1})U(z) = K_P(1 - z^{-1})E(z) + K_I T E(z) + \frac{K_D}{T}(1 - z^{-1})^2 E(z)$$

By applying inverse z-transform and by simplification yields:

$$u(k) = u(k-1) + K_P(e(k) - e(k-1)) + K_I T e(k) + \frac{K_D}{T}(e(k) - 2e(k-1) + e(k-2)) \quad (6.62)$$

The digital PID controller given by (6.62) is different in structure to that obtained from time-domain considerations and is known as the *velocity PID* algorithm.

6.4.3 Other digital controllers. Example

A digital controller implemented following various different z-transform methods.

A plant modeled by the continuous transfer function:

$$H_P(s) = \frac{Y(s)}{U(s)} = \frac{10}{s^3 + 7s^2 + 6s} \quad (6.63)$$

is to be controlled in the closed-loop with unity negative feedback, using a forward path lead compensator of transfer function

$$D(s) = \frac{1.5(s+1)}{s+3} \quad (6.64)$$

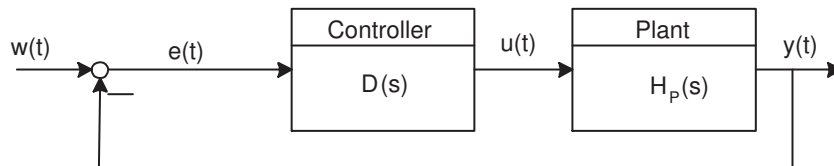


Figure 6.28: A closed-loop control system

The controller is to be implemented in digital form. We shall investigate the performance of implementations having a sampling period of 0.1s and being converted into the z-domain by:

- a) The simple method
- b) The Tustin method
- c) The z-transform method

Evaluating the controller transfer functions:

- a) Using the transformation:

$$s = \frac{1 - z^{-1}}{T}$$

with $T=0.1s$ gives

$$s = 10(1 - z^{-1}) \quad (6.65)$$

and so

$$D(z) = \frac{1.5[10(1 - z^{-1}) + 1]}{10(1 - z^{-1}) + 3} = \frac{16.6 - 15z^{-1}}{13 - 10z^{-1}} = \frac{1.2692 - 1.1538z^{-1}}{1 - 0.7692z^{-1}} \quad (6.66)$$

- b) Using Tustin substitution for s:

$$s = \frac{2(1 - z^{-1})}{T(1 + z^{-1})} \quad (6.67)$$

the transfer function converts to:

$$D(z) = \frac{1.5 \left[\frac{2(1 - z^{-1})}{T(1 + z^{-1})} + 1 \right]}{\left[\frac{2(1 - z^{-1})}{T(1 + z^{-1})} + 3 \right]} = \frac{1.3696 - 1.2391z^{-1}}{1 - 0.7391z^{-1}} \quad (6.68)$$

- c) Using the z-transform method

$$D(z) = Z \left(\frac{1 - e^{-sT}}{s} G(s) \right) = (1 - z^{-1}) Z \left(\frac{1.5(s + 1)}{s(s + 3)} \right) = \frac{1.50 - 1.3704z^{-1}}{1 - 0.7408z^{-1}} \quad (6.69)$$

The step responses of the controlled systems may be compared by means of an appropriate computer package or by means of Matlab. Graphs of the step responses are shown in Figure 6.29 and the following table compares them for the maximum overshoot at the time at which it occurs.

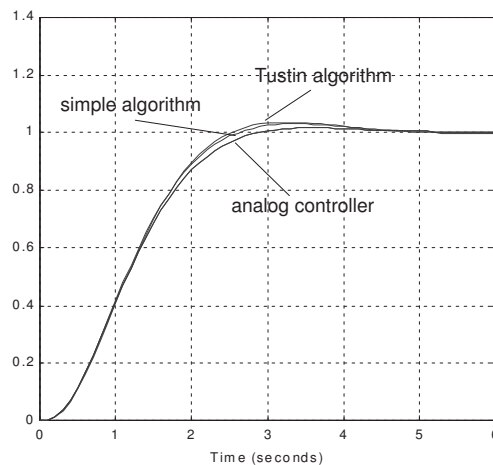


Figure 6.29: Closed-loop step responses

<i>Conversion type</i>	<i>simple</i>	<i>Tustin</i>
overshoot %	3.6	2.9
peak time, sec	3.2	3.35

For comparison, the analog controller would give an overshoot of 1.6% at 3.5 seconds, so all the digital implementations produce some degree of performance degradation.

In order to produce a digital control system arrangement for one of the above transfer functions in z , it will be necessary to produce a suitable program for the processor. A first step in doing so is to convert the transfer function into a discrete time (difference) equation from which an algorithm or pseudo-code may be developed.

The compensator transfer function in z is, by definition, equal to the ratio $U(z)/E(z)$ of the z -transform of the controller action u and the error signal e , respectively, thus:

$$D(z) = \frac{U(z)}{E(z)} = \frac{1.2692 - 1.1538z^{-1}}{1 - 0.7692z^{-1}}$$

which will give

$$(1 - 0.7692z^{-1})U(z) = (1.2692 - 1.1538z^{-1})E(z)$$

Such equation can be returned easily to the time domain by recalling that multiplying by z^{-1} represents a delaying of the associated signal by one sampling interval. For example $E(z)$ represents the latest sample of the error signal and $z^{-1}E(z)$ represents the sample at the previous sampling interval. Returning to the time-domain notation, it can therefore be seen that

$$u(k) - 0.7692u(k-1) = 1.2692e(k) - 1.1538e(k-1)$$

or

$$u(k) = 0.7692u(k-1) + 1.2692e(k) - 1.1538e(k-1) \quad (6.70)$$

The outline algorithm could then be:

Initialize: set $u(k-1) = 0$, $e(k-1) = 0$ or to their actual values if they are available

Loop: Reset sampling interval timer

Input e ($=e(k)$)

Calculate $u(k)$ from equation (6.70)

Output $u(k)$

Set $u(k-1) = u(k)$

Set $e(k-1) = e(k)$

Wait for end of sampling interval

End Loop

The "Wait for end of sampling interval" is because the calculations are unlikely to take the same exact time and, even they do, that time is unlikely to be the sampling interval. The easiest way of ensuring that sampling does take place at equal intervals of time maybe to program the "Loop" routine as an interrupt routine which is called by an appropriate pulse train (often generated by a support chip having a software-settable timing signal generator)

6.5 Exercises

6.5.1 Example 1: Pulse transfer function and system stability

Consider a first order system having the transfer function:

$$G(s) = \frac{1}{s + 4} \quad (6.71)$$

Compute the pulse transfer function and analyze the system stability.

Solution. We shall determine the pulse transfer function $G(z)$ using two methods (ZOH and simple substitution), as shown in Chapter 6 (Digital control systems).

Using the ZOH method, $G(z)$ is obtained from:

$$G_1(z) = Z\{G_0(s)G(s)\} = Z\left\{\frac{1 - e^{-sT}}{s} \frac{1}{s + 4}\right\} = (1 - z^{-1}) \frac{1}{4} Z\left\{\frac{1}{s} - \frac{1}{s + 4}\right\} \quad (6.72)$$

Using a table of the z-transforms of basic signals we obtain:

$$G_1(z) = \frac{z - 1}{z} \frac{1}{4} \left(\frac{z}{z - 1} - \frac{z}{z - e^{-4T}} \right) = \frac{1}{4} \frac{1 - e^{-4T}}{z - e^{-4T}} \quad (6.73)$$

Using the simple substitution $s = \frac{1-z^{-1}}{T}$, the pulse transfer function is:

$$G_2(z) = \frac{1}{\frac{1-z^{-1}}{T} + 4} = \frac{T}{1 + 4T - z^{-1}} = \frac{Tz}{(1 + 4T)z - 1} \quad (6.74)$$

For $T = 0.1$, the pulse transfer functions (written in z instead of z^{-1}) in all cases above are:

$$G_1(z) = \frac{0.08242}{z - 0.6703}, \quad G_2(z) = \frac{0.1z}{1.4z - 1} \quad (6.75)$$

The continuous system (6.24) is clearly stable, since it has only one negative pole at -4 .

The discrete system is described by any of the pulse transfer functions $G_1(z)$, $G_2(z)$. Each of them has one pole that is located inside the unit circle:

$$G_1(z): z_1 = 0.6703 < 1, \quad G_2(z): z_2 = \frac{1}{1.4} < 1 \quad (6.76)$$

thus the discrete system is stable, for both methods.

Moreover, if we analyze the general form of the pulse transfer functions (6.73) for ZOH and (6.74) for the simple substitution, the poles are smaller than 1 for any value of the sampling period T :

$$G_1(z): z_1 = e^{-4T} < 1, \quad \text{and} \quad G_2(z): z_2 = \frac{1}{1 + 4T} < 1, \quad \text{for any } T > 0 \quad (6.77)$$

6.5.2 Example: Pulse transfer function and system stability

Consider a critically stable system having the transfer function:

$$G(s) = \frac{1}{s^2 + 1}$$

The system has two poles on the imaginary axis, $s_{1,2} = \pm j$, thus it is critically stable.

We shall determine the pulse transfer function $G(z)$ and analyze the stability of the discrete system.

Solution. Using the ZOH method, the pulse transfer function result as:

$$G_1(z) = Z\{G_0(s)G(s)\} = Z\left\{\frac{1 - e^{-sT}}{s} \frac{1}{s^2 + 1}\right\} = (1 - z^{-1})Z\left\{\frac{1}{s} - \frac{s}{s^2 + 1}\right\} \quad (6.78)$$

Using a table of the z-transforms of basic signals (a step and a cosine) we obtain:

$$G_1(z) = \frac{z - 1}{z} \left(\frac{z}{z - 1} - \frac{z(z - \cos T)}{z^2 - 2z \cos T + 1} \right) \quad (6.79)$$

where the z-transform of the second term is obtained from:

$$\mathcal{L}\{\cos \omega t\} = \frac{s}{s^2 + \omega^2}$$

and

$$Z\{\cos \omega t\} = \frac{z(z - \cos \omega T)}{z^2 - 2z \cos \omega T + 1}, \quad \text{with } \omega = 1$$

Equation (6.79) can be written as:

$$G_1(z) = 1 - \frac{(z - 1)(z - \cos T)}{z^2 - 2z \cos T + 1} = \frac{z^2 - 2z \cos T + 1 - (z - 1)(z - \cos T)}{z^2 - 2z \cos T + 1} \quad (6.80)$$

The poles of the discrete system are the roots of the denominator:

$$z_{1,2} = \cos T \pm \sqrt{(\cos T)^2 - 1} = \cos T \pm j \sin T \quad (6.81)$$

The roots are complex and they are located exactly on the unit circle because the magnitude of the poles is unity: $|z_{1,2}| = \sqrt{(\cos T)^2 + (\sin T)^2} = 1$.

We shall compare this result with the poles of the discrete system resulted by simple transformation $s = (1 - z^{-1})/T = (z - 1)/zT$:

$$G_2(z) = \frac{1}{\left(\frac{z-1}{zT}\right)^2 + 1} = \frac{z^2 T^2}{(T^2 + 1)z^2 - 2z + 1} \quad (6.82)$$

with the poles:

$$z_{1,2} = \frac{1 \pm \sqrt{1 - (T^2 + 1)}}{T^2 + 1} = \frac{1 \pm Tj}{T^2 + 1} \quad (6.83)$$

It is clear that $z_{1,2}$ will have a magnitude of 1 only for $T = 0$. For other values of the sampling time, the simple transformation is an inaccurate approximation. If, for example $T = 0.1$:

$$z_{1,2} = \frac{1 \pm 0.1j}{0.1^2 + 1} = 0.9901 \pm 0.0990j, \quad \text{and } |z_{1,2}| = 0.995 < 1$$

For $T = 0.01$:

$$z_{1,2} = \frac{1 \pm 0.01j}{0.01^2 + 1} = 0.9999 \pm 0.0100j, \quad \text{and } |z_{1,2}| = 0.99995 < 1$$

The approximation is better when the sampling period decreases towards 0.

6.5.3 Example: Pulse transfer function from a difference equation

Consider a system with input u and output y whose behavior can be described with a difference equation:

$$y(k) + 2y(k-1) + \frac{3}{4}y(k-2) = u(k-1) \quad (6.84)$$

1. Define the pulse transfer function of the system
2. Is the system stable? Justify your answer.

Obs. $y(k) = y(kT)$, $y(k-1) = y((k-1)T)$ where T is the sampling period

Solution. If the initial conditions are considered zero, the following property of the Z transform is useful:

$$\text{Shift right: } Z\{f(k-n)T\} = z^{-n}F(z)$$

The z-transform of equation (6.84) is:

$$Y(z) + 2z^{-1}Y(z) + \frac{3}{4}z^{-2}Y(z) = z^{-1}U(z) \quad (6.85)$$

or

$$Y(z)(1 + 2z^{-1} + \frac{3}{4}z^{-2}) = z^{-1}U(z) \quad (6.86)$$

and the transfer function:

$$\frac{Y(z)}{U(z)} = \frac{z^{-1}}{1 + 2z^{-1} + \frac{3}{4}z^{-2}} = \frac{z}{z^2 + 2z + \frac{3}{4}} \quad (6.87)$$

The poles are:

$$z_{1,2} = -1 \pm \frac{1}{2}, \quad z_1 = -1.5, \quad z_2 = -0.5$$

One pole is located inside the unit circle while the other one is outside, therefore the system is unstable.

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